Muhammad Junaid Zaffar

Email: junaidzafar82@gmail.com

Research Interests

- Databases & Data Mining.
- Machine Learning & Deep Learning.
- AI for Healthcare (early detection, personalized care).

Academic History

Masters of Science (M.Sc) in Computer Science (CGPA 3.638/4.00)
(2017-2020) University of Engineering and Technology (UET), Lahore, Pakistan.
Thesis Title: Automatic Images Colorization Using Deep Learning.

Bachelor of Science (BS) in Computer Science (CGPA 3.54/4.00)
(2013 – 2017) The Islamia University of Bahawalpur, Pakistan.
FYP Title: Educational Management System-Integrated with Biometric.

Projects

- Hybrid Deep Learning for Predictive Anomaly Detection in Healthcare IoT (Ongoing).
- Towards Early and Accurate Detection of Pancreatic Cancer Using Artificial Intelligence (Ongoing).
- EmoSense - A Mobile Application for the Detection of Crowd Emotions Using Deep Learning (FYP Supervised).
- Cotton Help - A Mobile Application for the Detection of Cotton Leaf Diseases Using Deep Learning (FYP Supervised).
- Early detection of Prostate Cancer with the Help of Deep Learning (Paused).
- Detection of Tomato Plant Diseases using Deep Learning (FYP Supervised).
- Automatic Images Colorization Using Deep Learning (Master's thesis).
- Automatic Game Playing Using CNN Model (Semester Course Project).
- Vision-Based Human Activity Detection Using Deep Learning Model (Lab Assignment Project).
- Sentimental Analysis on the Basis of Tweets (Freelance Project).
- Absentee Predictions Based On Traditional Machine Learning Algorithms (Course Project).
- Branch Prediction for Multiple Instruction Issue (MII) Processors Using Machine Learning Techniques (Course Project).
- Head Node and Shake Detection (Internship Project).
- Educational Management System Integrated with Biometric (Bachelor's FYP).
- MLC Report System (Desktop application for Govt. THQ LODHRAN HOSPITAL).
- Manufacturing Management System (Desktop Application for Whizz Systems).
- SmartDR. (Desktop Application for a Client).
- Point of Sale (Website, Desktop, and Hybrid Mobile Application for a Client).

Work History

Lecturer – Department of Data Science, The Islamia University of Bahawalpur, Pakistan
(October 04, 2022 – To Date)
Responsibilities

- Teaching and designing core and elective courses to Undergraduate Students.
- Supervising final-year projects and actively engaging in research projects.

Courses Taught:

1. DSCS-111 Programming Fundamentals (Fall 23, Spring 24, Fall 24, Spring 25, Fall 25, Spring 26, Fall 26).
2. INFT-3103 Web Systems and Technologies (Fall 22, Spring 23, Fall 23, Fall 25).
3. DASC-6100 Big Data Analytics (Fall 22).
4. COSC-1213 Computing and Ethics (Fall 22, Summer 23).
5. DASC-3101 Object-Oriented Programming (Spring 23, Fall 24, Spring 26).
6. DASC-2116 Image Processing and Analysis (Spring 23, Spring 24, Fall 24, Fall 25, Fall 26).
7. DASC-5101 Operating Systems (Fall 24)

Lecturer – Faculty of Information and Technology, University of Central Punjab, Lahore
(October 05, 2020 – October 03, 2022)
Responsibilities

- Teaching and designing core and elective courses to Undergraduate Students.
- Supervising final-year projects and actively engaging in research projects.

Courses Taught:

1. CSCP1013 Programming Fundamentals (Spring 22, Summer 22).
2. CSCS1513 Introduction to Computing (Fall 20, Fall 21, Spring 22).
3. CSAL3203 Introduction to Image Processing (Fall 20, Spring 21, Summer 21).
4. CSSE4123 Software Quality Assurance (Spring 21).

Graduate Assistant – Department of Computer Science and Engineering, UET Lahore
(November 21, 2017 – January 22, 2020)
Responsibilities

- Teaching and designing laboratory subjects.
  - o Database Management System (Spring 18, Fall 18, Spring 19).
  - o Introduction to Computing (Fall 17, Fall 19).
  - o Programming Fundamentals (Spring 18, Fall 18, Spring 19).
- Assist respected faculty members in marking quizzes/assignments.
- Individual and/or group assistance to students in their final year and class projects.
- Member of OBE & Tutorial Group Committee.

Research Assistant – Computer Vision and Machine Learning Lab, KICS UET Lahore
(September 2017 – November 20th, 2017)
Duties

- Supported academic research in computer vision and machine learning.
- Conducted literature reviews, data collection, and analysis.
- Assisted in manuscript preparation and academic journal production.
- Co-led research planning, evaluation tools, and progress reporting with supervisor.

Junior Software Engineer – Concepts Solution, Bahawalpur
(February 2017 – June 2017)
Working in a small team on a variety of development projects and delivering solutions to meet and exceed clients' briefs.
Duties

- Design and develop Windows applications using C# (Windows Form).
- Analysing and debugging code.
- Maintaining the stability of all live systems.
- Outlining and presenting specifications for all software requirements.

Technical Skills

- **Programming Languages:** Python, C/C++, C#, JavaScript, Lua.
- **Machine Learning & Deep Learning:** TensorFlow, Keras, PyTorch, Scikit-learn, scikit-image, NumPy, Matplotlib, Pandas.
- **Data Science & Big Data:** PySpark, Hadoop, PowerBI.
- **Computer Vision & Image Processing:** OpenCV, Labelme, LabelImg, MakeSense.ai.
- **Databases & Reporting:** SQL (T-SQL, MySQL), NoSQL (MongoDB, Firebase), SQL Server Management Studio, SAP Crystal Reports.
- **Development Environments & Frameworks:** PyCharm, Visual Studio, VS Code, Sublime Text, Flutter, React Native, Octave.

Awards & Scholarships

- Graduate Assistantship – Served in the CSE Department at UET Lahore for over 2 years during my Master's studies.
- PEEF Graduate Merit Scholarship – Awarded for academic excellence during Master's.
- Position Holder Scholarship – Received 50% fee return across five semesters of undergraduate study (1st, 2nd, 3rd, 7th, 8th).

Training Workshops

- Attended training workshop on "Advancing Teaching Excellence: Strategies for Effective Higher Education Instruction and Professional Development" in DEC 12-14, 2023.
- Attended training workshop on "Enhancing Research Productivity" on SEPT 21, 2021.
- Attended training workshop on "Teaching Excellence Methodologies" on August 2021.
- Attended training workshop on "Outcome Based Teaching Practices" in 2018.

English Proficiency

- Duolingo English Test (120) **(**https://certs.duolingo.com/7569z57g2v6rjfwf).

Community Services

- Active Volunteer Member of a Blood Donor Society (Hope of Life) since 2015.
    - Activities involved are: reaching out to blood donors and arranging blood for the Cancer and Thalassemia Patients.
- Weekly free counseling for deserving students; programming problem-solving classes conducted twice monthly.

References

- **Dr. Muhammad Usman Ghani Khan**, Professor,
  University of Engineering and Technology, Lahore. **Email**: usman.ghani@uet.edu.pk
- **Dr. Syed Khaldoon Khurshid**, Associate Professor,
  University of Engineering and Technology, Lahore. **Email:** khaldoon@uet.edu.pk
- **Dr. Muhammad Ateeq**, Lecturer (Assistant Professor),
  University of Leicester, United Kingdom. **Email:** ma1133@le.ac.uk